

Breaking the 7/5 Barrier for Graphic TSP on Graphs of Girth at Least Five

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Abstract

We prove a polynomial-time $101/73$ approximation for the shortest-path metric of a connected unweighted graph containing no cycles of length three or four. Our proofs combine a minimum binary even core, whole-component vertex copying, an exact weighted disjoint-terminal-path optimization, a globally compatible fractional matching comparison, and connectivity-decreasing local switches on cycles of lengths five through twelve. The final calculation combines two constructible tours with weights $60/73$ and $13/73$.

1 Introduction

1.1 Problem, motivation, and main result

Let $G = (V, E)$ be a finite connected simple undirected graph with unit edge lengths, and put $n = |V|$. A *graphic tour* is a closed walk in G that visits every vertex, and $\text{OPT}(G)$ denotes its minimum number of edge traversals. Equivalently, $\text{OPT}(G)$ is the minimum number of edges, counted with multiplicity, in a connected spanning Eulerian multigraph obtained from copies of edges of G . For $n \geq 3$, this is also the length of a shortest Hamilton cycle in the complete metric defined by the shortest-path distances d_G : expand metric edges into shortest paths in one direction, and shortcut an Euler tour in the other. Edge multiplicities are allowed in a tour, but the input graph is simple.

Graphic TSP is a central meeting point of metric approximation, parity correction, connectivity augmentation, and structural graph theory. It is strictly more structured than arbitrary metric TSP—all distances arise from unit graph paths—but it remains APX-hard even on cubic bipartite graphs [7]. The difficulty is the tension between cheap parity and cheap connectivity. A sparse even subgraph can have many components; a well-connected subgraph can be expensive to make Eulerian. Most successful algorithms balance two or more certificates that expose these costs in different ways.

This paper studies a different structural restriction: the input has *girth at least five*, so it contains neither a triangle nor a four-cycle. We use the standard convention that a forest has girth $+\infty$. The hypothesis imposes neither a maximum-degree bound nor a planarity assumption, and it permits cut vertices and bridges; it is therefore incomparable with bounded-degree assumptions. The restriction is useful algorithmically because every nontrivial even component contains only cycles of length at least five, short even components admit a finite collection of cycle states, and a lexicographically minimum even core satisfies the strict density inequality proved in Lemma 2.6. Our main result is the following.

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Theorem 1.1 (Main theorem). *For every connected unweighted simple graph G of girth at least five, the deterministic algorithm described in Section 5.2 produces, in polynomial time, a graphic tour with*

$$\text{ALG}(G) \leq \frac{101}{73} \text{OPT}(G).$$

Forests are permitted and are handled exactly. There is no restriction on the maximum degree.

Numerically,

$$\frac{101}{73} = 1.3835616438\dots, \quad \frac{7}{5} - \frac{101}{73} = \frac{6}{365} = 0.0164383561\dots$$

Thus the theorem improves on the general Graphic-TSP benchmark on this class.

The only finite case analyses concern cycle bags of orders at most twelve. Lemma 4.4 states their exact domains, and Appendix A gives human-checkable proofs. The reduction of arbitrary input graphs to those finite domains is proved mathematically.

1.2 Related work

The classical Christofides–Serdyukov method combines a minimum spanning tree with a minimum parity correction and gives a $3/2$ approximation for metric TSP [2, 16]. Karlin, Klein, and Oveis Gharan later obtained a randomized $(3/2 - \varepsilon)$ approximation for general metric TSP, for an explicit constant $\varepsilon > 10^{-36}$ [10]. For Graphic TSP, Oveis Gharan, Saberi, and Singh obtained a $(3/2 - \varepsilon_0)$ bound using randomized spanning-tree rounding [14]. Mömke and Svensson introduced the adding-and-removing-edges framework and obtained 1.461 [11]; Mucha sharpened its analysis to 13/9 [12]. Sebő and Vygen then combined optimized ear decompositions, earmuffs, and parity correction to obtain a $7/5$ approximation [15], the general benchmark recorded in the monograph of Traub and Vygen [17]. A manuscript claiming $4/3$ for general Graphic TSP was later withdrawn [3], so we do not treat that factor as established.

Stronger guarantees are known under degree restrictions. Results of Mömke–Svensson and Boyd, Sitters, van der Ster, and Stougie give $4/3$ for subcubic graphs [11, 1]; Correa, Larré, and Soto improve slightly on $4/3$ for cubic graphs [4]; Dvořák, Král, and Mohar give $9/7$ for cubic graphs [5]; and Newman gives $25/18$ for subquartic graphs [13]. Wigal, Yoo, and Yu prove the sharper tour bound $(5n + n_2)/4 - 1$ for simple two-connected subcubic graphs, yielding $5/4$ in the cubic case [18]. These results are stronger on some intersections but do not cover girth-at-least-five graphs of unrestricted degree. Conversely, our theorem does not cover bounded-degree graphs with triangles or four-cycles.

The subtour relaxation remains a principal lower bound, with conjectured metric-TSP integrality gap $4/3$ [11, 13, 17]. Our analysis instead uses the integral potential $\Phi \leq \text{OPT}$ and therefore makes no LP-gap claim. Graphic TSP is APX-hard on cubic bipartite graphs [7], but those graphs may contain four-cycles. For arbitrary-degree graphs of girth at least five, the general $7/5$ algorithm of Sebő–Vygen is therefore the directly applicable comparison benchmark.

1.3 Contributions and proof strategy

Our proof has four main steps.

- (i) For every connected input of girth at least five that contains a cycle, a lexicographically minimum *binary even core* F , computable by perfect matching, satisfies $n \leq \Phi(F) = |F| + 2i(F) \leq \text{OPT}$ and $f_C \leq 2v_C - 6$ on each noncycle nontrivial component C of F .

- (ii) Whole-component activation and exact zero-distance copying replace each selected core component by vertex-disjoint cycle bags of orders five through twelve. Unselected components remain in a global debit; the transformation preserves the comparator optimum and permits cost-nonincreasing projection back to G .
- (iii) Complementary local covers, a circuit inequality, and one exact prescribed-cardinality disjoint- $\mathcal{S}$ -path instance jointly choose interbag paths, outside coverage, and connectivity-paying switches.
- (iv) A globally compatible fractional perfect matching compares this paid construction with an optimum tour. Blending it with the direct core tour in weights $60/73$ and $13/73$ yields $101/73$; all component rows except C_5 and C_{11} retain explicit slack.

Our proof balances two constructible certificates. The direct certificate connects a lexicographically minimum even core by doubled edges. The paid certificate activates its short components, uses an exact disjoint-path oracle to connect their cycle bags, and applies local switches to reduce the remaining connectivity cost. An optimum metric Hamilton cycle is used only to compare with the oracle: a fractional perfect-matching point turns its crossing segments into feasible path solutions, while the girth hypothesis charges every bad local transition to metric excess or a return vertex. With $\rho = 16/13$, the blend $(60P + 13D)/73$ and $\Phi \leq \text{OPT}$ give $101 \text{ OPT} / 73$.

The only black boxes are minimum-weight perfect matching and Edmonds's perfect-matching polytope [6], together with Yamaguchi's prescribed-cardinality disjoint- $\mathcal{S}$ -path reduction and deterministic weighted linear matroid parity [19, 9]. Sections 2–5 prove all reductions and hypotheses; Appendix A gives the bounded local certificates.

2 Core, activation, and the direct tour

2.1 The even core and direct completion

Definition 2.1. A *binary even core* is an edge subset $F \subseteq E$ such that every vertex has even degree in (V, F) . Isolated vertices are allowed. Let $i(F)$ denote their number, let $\text{cc}(V, F)$ denote the number of connected components of (V, F) (including isolated vertices), and set

$$\Phi(F) = |F| + 2i(F).$$

Throughout the paper, F denotes a binary even core minimizing $(\Phi(F), |F|)$ lexicographically.

The proofs of the core optimization and lower bound are given in Section 2.2. The properties used in the main argument are

$$\Phi(F) \leq \text{OPT}(G), \quad v_C \leq f_C \leq 2v_C - 6 \quad \text{for every noncycle nontrivial component } C \text{ of } (V, F). \quad (1)$$

The first inequality is used only when the connected input contains a cycle; trees are handled separately. Here $v_C = |V(C)|$ and $f_C = |E(C)|$.

Lemma 2.2 (Doubled-quotient completion). *Let H be an even spanning multigraph on V whose edges are copies of edges of the connected input graph G . If H has c connected components, including isolated vertices, then it can be made connected by adding at most $2(c - 1)$ copies of edges of G .*

Proof. Contract the components of H in G and delete the resulting loops. The quotient is connected, so it has a spanning tree with $c - 1$ edges. Add two copies of the corresponding original edge of G for every tree edge. These additions connect the components of H and change every degree by an even number. Parallel edges in the quotient cause no difficulty. $\square$

For the binary core this produces the first tour of cost

$$D \leq |F| + 2(\text{cc}(V, F) - 1). \quad (2)$$

The core is a lower-bound object and an algorithmic object, not a restriction on the edges allowed to an optimum tour.

2.2 Core optimization and structural bounds

Matching reduction. Write $n = |V|$, $m = |E|$, and $M = m + 1$. The integer objective

$$J(F) = M\Phi(F) + |F| = (M + 1)|F| + 2Mi(F)$$

implements the lexicographic minimization exactly: a one-unit change in Φ dominates every possible change in $|F| \in [0, m]$, because $M = m + 1$ is larger than the full range of the second objective.

At a vertex v of input degree d_v , create one port $p_{v,e}$ for each incident edge and d_v slots $q_{v,1}, \dots, q_{v,d_v}$. Join every port to every slot at zero cost. Make the slots a zero-cost clique, except that for $d_v \geq 2$ the distinguished edge $q_{v,1}q_{v,2}$ has cost $-2M$. For each original edge $e = uv$, join $p_{u,e}$ to $p_{v,e}$ with cost $M + 1$. The auxiliary graph has a perfect matching, obtained by matching all ports to their slots.

Lemma 2.3 (Core optimization). *A minimum-weight perfect matching of this auxiliary graph decodes to a lexicographically minimum binary even core. The construction has $4m$ vertices, $O(m^2)$ edges, and integer costs with $O(\log(m + 2))$ bits, and is polynomial-time solvable.*

Proof. Decode F as the set of original edges whose external port edges are matched. If t_v such external edges are incident with the ports at v , there remain $d_v - t_v$ unmatched local ports, each of which must consume a slot. Exactly t_v slots remain to be matched among themselves, so t_v is even. Thus every perfect matching decodes to a binary even core.

If $t_v = 0$, no slot–slot edge is used; in particular the negative distinguished edge cannot collect a reward at an isolated vertex. If $t_v > 0$, the local reward is at most $2M$. Hence every matching decoding to F has cost at least

$$(M + 1)|F| - 2M(n - i(F)) = J(F) - 2Mn.$$

Conversely, for any binary even F , at an active vertex ($t_v > 0$) reserve slots 1 and 2 for their negative pair, reserve any further $t_v - 2$ slots and pair those reserved slots arbitrarily among themselves, and match the other slots bijectively to the remaining ports. At an inactive vertex use only port–slot edges. This realizes equality. Consequently

$$\min\{\text{matching cost}\} + 2Mn = \min_F J(F),$$

and a minimum matching gives the desired core.

The exact auxiliary vertex count is $2\sum_v d_v = 4m$. The edge count is $m + \sum_v (d_v^2 + \binom{d_v}{2}) = (3/2)\sum_v d_v^2 = O(m^2)$. Together with the stated cost encoding, polynomial-time solvability now follows from weighted perfect matching [6, 8]. $\square$

Tour identity and lower bound.

Lemma 2.4 (Binary-core tour identity). *For a connected graph on at least two vertices,*

$$\text{OPT}(G) = \min_{\substack{Q \subseteq E: \\ (V, Q) \text{ is even}}} (|Q| + 2(\text{cc}(V, Q) - 1)).$$

Proof. For every admissible Q , Lemma 2.2 applied to (V, Q) constructs a tour of the displayed cost. For the other direction, take an optimum connected spanning even multigraph T . No edge has multiplicity more than two: removing two copies from an edge with at least three preserves support connectivity and parity while decreasing the cost. Let F_* be the set of underlying edges having multiplicity one in T , and let D_* be the set having multiplicity two. Reducing degrees modulo two shows that F_* is binary even. Every edge of F_* lies within one component of (V, F_*) ; because T is connected, the edges of D_* therefore connect the quotient obtained by contracting those components. Hence $|D_*| \geq \text{cc}(V, F_*) - 1$, and

$$\text{OPT} = |F_*| + 2|D_*| \geq |F_*| + 2(\text{cc}(V, F_*) - 1).$$

□

The identity does not give an efficient minimization of the component penalty. The matching reduction optimizes Φ , not that penalty.

Lemma 2.5 (Core lower bound). *If G is connected and contains a cycle, then the minimum core is nonempty and*

$$|V| \leq \Phi(F) \leq \text{OPT}(G).$$

Proof. A cycle Z of length $\ell \geq 3$, together with a doubled spanning tree after Z is contracted, gives a tour of cost $\ell + 2(n - \ell) = 2n - \ell < 2n - 2$. Thus the odd support F_* of an optimum tour in the preceding proof cannot be empty: an all-doubled connected support costs at least $2n - 2$. Since F_* has a nontrivial component, $i(F_*) \leq \text{cc}(V, F_*) - 1$. Consequently

$$\Phi(F) \leq \Phi(F_*) \leq |F_*| + 2(\text{cc}(V, F_*) - 1) \leq \text{OPT}.$$

Every nonisolated vertex of (V, F) has positive even degree and hence degree at least two. The degree sum therefore gives $|F| \geq n - i(F)$, so $\Phi(F) \geq n + i(F) \geq n$. The minimum core is nonempty also directly: retaining the cycle and isolating other vertices has $\Phi = 2n - \ell < 2n$, the empty-core value. □

For a tree the only binary even core is empty: every nonempty finite even subgraph contains a cycle. Thus $\Phi = 2n$, whereas every tree edge is a bridge and must be traversed in both directions, giving $\text{OPT} = 2n - 2$. This is why the tree branch is necessary.

Lexicographic density improvement.

Lemma 2.6 (Core component density). *In a girth-at-least-five input, each noncycle nontrivial component of a lexicographically minimum core satisfies $v_C \leq f_C \leq 2v_C - 6$. The same inequality holds for a simple cycle component of length at least six.*

Proof. Every vertex of the nontrivial even component C has degree at least two, so the degree sum gives $v_C \leq f_C$. Retain a simple cycle Z of length $\ell \geq 5$ in C , delete the other selected edges of C , and thereby isolate its other vertices. The old contribution to Φ is f_C and the new one is $\ell + 2(v_C - \ell) = 2v_C - \ell$. Minimality gives $f_C \leq 2v_C - \ell$. If C is not a cycle and equality holds, then $f_C > \ell$, so the replacement decreases the number of selected edges strictly while preserving Φ , contrary to the tie-break. Hence $f_C \leq 2v_C - \ell - 1 \leq 2v_C - 6$. For a cycle of length at least six the claimed inequality is immediate. A pentagon is the exception and is deliberately activated. □

Remark 2.7. The argument uses neither a spanning 2-factor nor an LP comparison: minimum binary even cores can contain vertices of degree greater than two, and only the proved inequality $\Phi \leq \text{OPT}$ is invoked below. Thus the theorem is an approximation relative to the optimum tour, not an integrality-gap statement.

2.3 Activation and zero-distance copies

For the remainder of the main construction, assume that G contains a cycle; the tree case has already been separated. In particular, Lemma 2.5 applies to the chosen core F .

Select every simple cycle component of F of length $5, \dots, 12$, and every noncycle component having at most seventeen edges. Decompose each selected component into edge-disjoint simple cycles. Such a decomposition is found by repeatedly removing the edges of a simple cycle: all remaining degrees stay even, so the process continues in every nontrivial remaining component until no edge remains.

Lemma 2.8 (Activation bounds). *Every cycle in these decompositions has length between five and twelve. The unselected nontrivial components are either simple cycles of length at least thirteen, or noncycle components with at least eighteen edges.*

Proof. Every cycle has at least five edges by the girth hypothesis. A noncycle even connected component has at least two cycles in any edge-disjoint cycle decomposition; otherwise all its edges would form one simple cycle. If its total edge count is at most seventeen, such a decomposition has at most three cycles, since four cycles would use at least twenty edges. Thus it has two or three cycles. With two cycles, each has length at most $17 - 5 = 12$; with three, each has length at most $17 - 10 = 7$. A selected simple-cycle component already has length between five and twelve. The description of the unselected components follows directly from the selection rule. $\square$

Replace each occurrence of an original vertex in a selected cycle by a separate formal copy. Thus the selected cycles become vertex-disjoint *bags*. Let A be the set of their formal vertices, B the unchanged vertices outside all selected components, and $\pi : A \cup B \rightarrow V$ the projection. Put

$$d'(u, v) = d_G(\pi(u), \pi(v)), \quad r_k = \#\{C_k \text{ bags}\}, \quad N_k = kr_k.$$

We regard d' as the edge-length function of the complete formal pseudometric on $A \cup B$; distinct copies of one original vertex may be at distance zero. Put $n' = |A| + |B|$. Let f and v be the edge and vertex totals in unselected nontrivial components, and let $i = i(F)$. Then

$$|A| = \sum_{k=5}^{12} N_k, \quad |B| = v + i, \quad \Phi = \sum_{k=5}^{12} N_k + f + 2i, \quad n' = |A| + |B| \leq \Phi \leq \text{OPT}. \quad (3)$$

Indeed, the cycle decompositions partition all edges of the selected components, so their total number of edges is $|A| = \sum_{k=5}^{12} N_k$. Whole-component activation leaves precisely the v vertices of the unselected nontrivial components and the i core isolates in B . Finally, every unselected nontrivial component satisfies $v_C \leq f_C \leq 2v_C - 6$ by Lemma 2.6, so

$$n' = \sum_{k=5}^{12} N_k + v + i \leq \sum_{k=5}^{12} N_k + f + 2i = \Phi;$$

the last inequality in (3) is Lemma 2.5.

Lemma 2.9 (Exact copying and projection). *The optimum Hamilton-cycle cost in the complete formal pseudometric is exactly $\text{OPT}(G)$. Every pair of distinct formal vertices at distance zero consists of two vertices of A in different bags. Any even spanning multigraph on the formal vertices can be projected and expanded into a spanning even multigraph on G without increasing either its cost or its number of connected components.*

Proof. Take an optimum metric Hamilton cycle on the original vertices. Insert every extra copy consecutively next to the occurrence of its original vertex. If there are g extra copies, this adds exactly g zero edges and changes no positive edge cost. Hence the formal optimum is at most $\text{OPT}(G)$. A cycle in a simple-cycle decomposition never contains an original vertex twice. Thus different copies of the same vertex belong to different active bags. No original vertex has both an active copy and an outside copy, because entire components were selected. Since $d_G(x, y) = 0$ only when $x = y$, this also proves the assertion about zero-distance pairs.

For projection, identify copies of each original vertex. The sum of their degrees is even. A zero edge becomes a loop; deleting it subtracts two from that sum. Expand every positive metric edge into an original shortest path. Each interior occurrence adds two to its degree. Every original vertex has at least one formal representative, because the selected cycle decompositions cover every edge, and hence every vertex, of each selected nontrivial component. The projected and expanded multigraph is therefore spanning and even. Identification can only merge components, deletion of a resulting loop cannot split one, and replacing an edge by a path cannot increase the component count. Costs are preserved under shortest-path expansion and do not increase when zero loops are deleted.

In particular, projecting any formal Hamilton cycle yields a graphic tour in G of no greater cost. Thus the formal optimum is at least $\text{OPT}(G)$, which, together with the lifted optimum constructed above, proves equality. $\square$

Remark 2.10. Activating just part of a component could leave both an active and an outside copy of the same vertex. This would create a zero active–outside distance and break the nonnegative-length reduction below. No such partial activation is used.

Metric-edge convention. Until the final projection, every edge of the complete formal pseudometric is treated as one atomic formal edge. When such an edge is expanded into a shortest path in G , every internal occurrence on that path is regarded as a private copy, even if its projection is also represented elsewhere in the formal construction. These private occurrences are identified with their original vertices only during final projection. Thus metric-edge expansion preserves formal component accounting and parity, and it can only merge components after projection. In particular, an original vertex encountered only inside an expanded metric edge is not counted as an explicitly used outside vertex in the variable z introduced below.

Applying (2) to the *original* core, and overcounting a selected component’s one connector charge by one charge per decomposed cycle, gives

$$D \leq \sum_{k=5}^{12} \left(1 + \frac{2}{k}\right) N_k + \frac{15}{13}f + 2i - 2. \quad (4)$$

To see this explicitly, let q be the number of unselected nontrivial components. The selected edges contribute $\sum_k N_k$, and replacing the two-unit connector allowance of each selected component by one allowance for each of its bags contributes at most $2 \sum_k r_k$. Every unselected nontrivial component has at least thirteen edges by Lemma 2.8, so $2q \leq 2f/13$. The i isolated vertices contribute $2i$, and the global quotient-tree saving contributes -2 . Substitution gives (4).

3 The paid tour construction

3.1 Ports and complementary covers

A path chosen between two different bags exposes one *port* at each endpoint. The paths used below are vertex-disjoint, so a formal bag vertex is exposed at most once. A feasible port set has even cardinality in every bag. A *local cover* of a bag with port set X is a spanning multigraph on that bag, possibly using private internal vertices as described in the rescue construction below, whose odd-degree vertices are exactly X . Its connected components are called its *pieces*. When every piece contains two ports, the pieces induce a perfect pairing of X .

Consider first a unit cycle of length k with $2j > 0$ ports in cyclic order. Let $\ell_0, \dots, \ell_{2j-1}$ be the positive lengths of its successive port-to-port arcs, with indices taken modulo $2j$, and put

$$a = \min \left\{ \sum_{t \text{ even}} \ell_t, \sum_{t \text{ odd}} \ell_t \right\}.$$

Lemma 3.1 (Two complementary covers). *There are spanning local covers with j vertex-disjoint pieces, each piece having exactly two odd vertices, its ports, and with costs*

$$L = k + a - 2j, \quad H = 2k - a - 2j. \quad (5)$$

Their two port pairings have union a single alternating cycle. With two ports at cyclic distance $d \leq k/2$, a connected spanning odd-endpoint walk of cost $k + d - 2$ exists. With no ports, retain the cycle.

Proof. Choose one of the two alternating sets of port-to-port arcs and keep every edge of its arcs once. On each intervening arc, omit one edge and take two copies of every other edge. The omitted edge splits that intervening arc into two doubled subpaths (either may be empty), one attached to each neighboring kept arc. Hence the resulting j pieces are connected, vertex-disjoint, and jointly contain every cycle vertex. Every nonport has even degree, whereas each port has one incident kept edge and an even number of incident doubled edges. Thus the only odd vertices of a piece are the two endpoints of its kept arc.

If the kept arcs have total length s , the cost is $s + 2(k - s - j) = 2k - s - 2j$. Their two possible total lengths are a and $k - a$, which give H and L , respectively. Since $a \leq k/2$, $L \leq H$. Exchanging the alternating sets pairs each port with its other cyclic neighbor, so the union of the two pairings is the cyclic alternating circuit through all $2j$ ports.

For two ports, take the longer arc once; on the shorter arc, of length d , omit one edge and double the other $d - 1$ edges. Both doubled subpaths attach to the longer arc, so the result is connected and spanning, its only odd vertices are the two ports, and its cost is $(k - d) + 2(d - 1) = k + d - 2$. With no ports the original cycle is already a connected spanning even graph. $\square$

3.2 The circuit inequality and switching algorithm

For perfect pairings P, Q on the same $2j$ labels, let $c(P, Q)$ be the number of alternating cycles in their multigraph union, where one copy of a common pair comes from each pairing. Thus a common pair is a two-edge alternating cycle and is not canceled.

Lemma 3.2 (Circuit inequality). *If $c(P, Q) = 1$, then for every perfect pairing R ,*

$$c(R, P) + c(R, Q) \leq j + 1. \quad (6)$$

If a bag with complementary pairings P, Q currently touches t global components, switching its pairing can be arranged to leave at most $j + 1 - t$ components in place of those touched components. Thus the global component count decreases by at least $2t - j - 1$.

Proof. A two-pair switch deletes two pairs of a perfect pairing and inserts one of the other two pairings of their four endpoints. The minimum number of such switches needed to transform P into Q is $j - c(P, Q)$. Indeed, one switch changes the number of alternating cycles with Q by at most one, giving the lower bound. Conversely, if an alternating cycle has length at least four, a switch can insert one pair of Q and split off one more alternating cycle. Repeating this operation proves the matching upper bound. The triangle inequality for this switch distance gives

$$j - c(P, Q) \leq (j - c(P, R)) + (j - c(R, Q)),$$

which is (6).

For the graph assertion, contract every local piece in every bag and every chosen physical path to an edge. In the formal graph used for component counting, these objects have disjoint interiors (rescue occurrences are private), and each port is incident with one local edge and one path edge. The contracted graph is therefore a disjoint union of alternating cycles. Delete from it the P -edges of the bag under consideration. The resulting path segments pair the bag's ports; call this pairing R . Reinstating P shows that $c(R, P) = t$, because its alternating cycles are exactly the previous global components that met the bag. Reinstating Q instead creates $c(R, Q)$ formal components; later projection can only merge them. By (6), $c(R, Q) \leq j + 1 - t$. The number of components therefore drops by at least $t - (j + 1 - t) = 2t - j - 1$. Before and after the switch the uncontracted graph is even: at each port the odd local degree is complemented by its path endpoint, and every other degree is even. $\square$

Choose the cheaper cover initially. A state is *free* when $L = H$. Designate a nonfree state *paid* precisely when

$$\begin{aligned} k \in \{5, 7, 9, 11\}, \quad a = (k - 1)/2, \quad \text{or} \\ (k, j, a) \in \{(8, 3, 3), (10, 3, 4), (12, 3, 5), (12, 4, 5)\}. \end{aligned} \tag{7}$$

All remaining states are *ignored* and retain the cheaper cover. Throughout the switching procedure, components are counted in the formal multigraph before shortest-path expansion and projection. Use the following two phases:

1. Repeatedly perform a free switch whenever it strictly decreases the number of formal components.
2. Repeatedly inspect all free and all designated paid switches, in both directions. Accept one only if it strictly decreases the component count and its increase in edge cost is at most that decrease.

Both phases terminate after at most a linear number of accepted operations, because every accepted switch lowers the integer component count. The formal component count can be computed after each candidate switch, so the procedure is polynomial. Notice that a switch may rearrange the component partition; none of the accounting below assumes monotone refinement.

For a bag X at a given stage, let t_X be the number of global components containing a port-bearing local piece of X . Let c_0, c_1 be the numbers of global components containing ports immediately before and immediately after phase two, and let t_X^0, t_X^1 be the corresponding bag incidences. Every accepted phase-two switch affects only port-containing components. If its component-count decrease is q , its local cost increase is at most q ; summing over phase two gives

$$\Delta_{\text{loc}} \leq c_0 - c_1, \quad \Delta_{\text{loc}} + 2c_1 \leq c_0 + c_1. \tag{8}$$

Every port-containing component contains a chosen path, whose endpoints belong to two different bags. It is therefore incident with at least two bags, and double counting incidences yields

$$c_0 + c_1 \leq \frac{1}{2} \sum_X (t_X^0 + t_X^1). \quad (9)$$

The term $2c_1$ in (8) is the two-unit connector allowance for the final port-containing components; the single global two-unit saving is taken only at the end of the construction.

The following caps drive the local accounting. In a free $2j$ -port state, termination of either phase implies

$$t_X \leq \left\lfloor \frac{j+1}{2} \right\rfloor,$$

because a larger value makes $2t_X - j - 1 > 0$ and Lemma 3.2 supplies a free component-reducing switch. Thus this cap holds both before and after phase two. In a designated paid state, let $\delta = H - L \geq 1$. Initially $t_X^0 \leq j$, since the local cover has only j pieces. At termination,

$$t_X^1 \leq \left\lfloor \frac{j+\delta}{2} \right\rfloor.$$

Indeed, a larger integer value gives $2t_X^1 - j - 1 \geq \delta$; switching to the complementary cover would then strictly reduce the component count by at least its possible cost increase δ , contrary to termination. By (9), a free state is charged its common cap, a paid state is charged half the sum of its initial and final caps, and an ignored state is charged the trivial cap j at both stages.

3.3 Two-port rescue and shadow penalties

For $k = 6, 8, 10, 12$, the potentially bad pairs are opposite cycle vertices. For $k = 7, 9, 11$, they have cyclic distance $(k-1)/2$. Fix a bag and one such pair x, y . Call the pair *bad* if the original graph contains no $\pi(x) - \pi(y)$ walk of length at most the threshold h_k in Table 1 that visits the projection of every vertex of the bag. All other pairs are *nonbad*, and a witnessing walk is a *rescue*. This test is exact: run a shortest-path search in the unweighted product graph with states (u, S) , where $u \in V(G)$ and S is the subset of the bag's k projected vertices visited so far. A transition along $uv \in E(G)$ updates S by adding v when appropriate. There are at most $2^k|V| \leq 2^{12}|V|$ states and $O(2^k|E|)$ transitions per test.

For completeness, a rescue walk is lifted to the formal construction as follows. Regard its successive vertex occurrences as distinct. Use the two endpoint occurrences for x, y , choose one visited occurrence for every other formal vertex of the bag, and identify only those chosen occurrences with the corresponding formal vertices. Keep every remaining internal occurrence as a private copy. The lifted walk is connected and spans the bag; all its vertices except x, y have even degree. On final projection, all private copies are identified with their original vertices, which preserves parity and can only merge components. Private occurrences, including occurrences of vertices outside the bag, receive no outside-coverage reward.

In a two-port state, the algorithm uses the direct walk of Lemma 3.1, except that for a potentially bad pair it uses a shortest rescue whenever the product search finds one of length at most h_k . Thus every nonbad farthest pair is represented by a local walk of cost at most h_k , while a bad pair uses the direct cycle walk.

For C_5 , a good pair means an adjacent pair on the cycle. Girth five precludes any internal chord of a pentagon. A good pair has a spanning path of cost four; a nonadjacent pair has a spanning walk of cost five. Let a_5 count good two-port pentagons and b_5 four-port pentagons. For $k \geq 6$, let c_k count bad two-port bags.

k	baseline d_k	threshold h_k	shadow penalty w_k	λ_k	paid coefficient p_k
5	7	–	reward -1	$-2/5$	$49/50$
6	8	6	4	0	$14/15$
7	$19/2$	7	2	$4/5$	$69/70$
8	11	9	4	0	$39/40$
9	$25/2$	10	2	$4/5$	$91/90$
10	14	12	4	0	1
11	$31/2$	13	2	$4/5$	$113/110$
12	17	15	4	0	$61/60$

Table 1: Constants. The column λ_k , used in Section 4, is a comparison loss in score units, with a negative value denoting reward. The coefficient used in Section 5 is $p_k = (d_k + \lambda_k/4)/k - 2/5$.

Lemma 3.3 (Multiport accounting). *Under the free/paid/ignored classification above, every $2j$ -port state has total local, crossing, and incidence contribution at most the corresponding baseline d_k in Table 1.*

Proof. The $2j$ ports split the cycle into two alternating families of j positive-length arcs. If a is the smaller family total, then

$$5 \leq k \leq 12, \quad 2 \leq j \leq \lfloor k/2 \rfloor, \quad j \leq a \leq \lfloor k/2 \rfloor,$$

and every such triple occurs by composing a and $k - a$ into j positive parts and interlacing them. Lemma 3.1 gives

$$L = k + a - 2j, \quad H = 2k - a - 2j, \quad \delta = H - L = k - 2a.$$

The $2j$ ports contribute j crossing units. Hence the contributions of an ignored, free, and paid state are, respectively,

$$C_{\text{ign}} = L + 2j = k + a, \tag{10}$$

$$C_{\text{free}} = L + j + \left\lfloor \frac{j+1}{2} \right\rfloor, \tag{11}$$

$$C_{\text{paid}} = L + j + \frac{1}{2} \left(j + \left\lfloor \frac{j+\delta}{2} \right\rfloor \right). \tag{12}$$

If k is even, an ignored state has $a \leq k/2 - 1$, so $C_{\text{ign}} \leq 3k/2 - 1 = d_k$. If $k \geq 7$ is odd, every state with $a = (k-1)/2$ is paid; thus an ignored state has $a \leq (k-3)/2$ and $C_{\text{ign}} < (3k-2)/2 = d_k$. There is no ignored multiport state for $k = 5$.

A free state has even k and $a = k/2$. Since $\lfloor (j+1)/2 \rfloor \leq j-1$ for $j \geq 2$,

$$C_{\text{free}} = \frac{3k}{2} - j + \left\lfloor \frac{j+1}{2} \right\rfloor \leq \frac{3k}{2} - 1 = d_k.$$

For an odd paid state, $a = (k-1)/2$ and $\delta = 1$, whence

$$C_{\text{paid}} = \frac{3k-1}{2} - \frac{j}{2} + \frac{1}{2} \left\lfloor \frac{j+1}{2} \right\rfloor \leq \frac{3k-2}{2}.$$

This equals d_k for $k = 7, 9, 11$ and is $13/2 < d_5 = 7$ for $k = 5$. Finally, each even paid state in (7) has $a = k/2 - 1$ and $\delta = 2$. Using $\lfloor (j+2)/2 \rfloor \leq j$ gives

$$C_{\text{paid}} = \frac{3k}{2} - 1 - \frac{j}{2} + \frac{1}{2} \left\lfloor \frac{j+2}{2} \right\rfloor \leq \frac{3k}{2} - 1 = d_k.$$

These cases exhaust all ignored, free, and paid states. □

For a two-port state at cyclic distance d , Lemma 3.1 gives internal cost $k + d - 2$. Adding its one crossing unit and its one incidence unit gives $k + d$. At a farthest pair, a rescue lowers this to at most $h_k + 2$. Direct substitution in Table 1 now shows that every nonbad two-port state has contribution at most d_k , whereas a bad one has contribution at most $d_k + w_k/4$. The same substitution for the no-port value $k + 2$, together with Lemma 3.3, shows that every zero-port and multiport state is bounded by d_k . Consequently, for each $k \geq 6$, the total contribution of all C_k bags is at most

$$d_k r_k + \frac{w_k}{4} c_k. \quad (13)$$

For pentagons the contributions are 7 at zero ports, 6 at good two ports, 7 at nonadjacent two ports, and $13/2$ at four ports. Hence their exact aggregate upper bound is $7r_5 - a_5 - b_5/2$. We deliberately use the weaker, score-compatible bound

$$\text{pentagon contribution} \leq 7r_5 - \frac{1}{4}(a_5 + 2b_5). \quad (14)$$

3.4 Exact paid path selection

For each formal bag vertex $u \in A$, create a physical terminal p_u and a shadow terminal q_u , joined by a spoke. The *physical layer* consists of the terminals p_u and the vertices of B ; it is complete, and the edge between the physical representatives of formal vertices $u, v \in A \cup B$ has length $d'(u, v)$. For each bag, its shadow terminals form a clique. There are no other auxiliary edges. A pentagon shadow edge has unshifted weight -1 for adjacent cycle vertices and zero otherwise. For $k \geq 6$, its unshifted weight is w_k on bad pairs and zero on all other pairs.

All physical terminals p_u belonging to one bag form a single terminal class, while every shadow terminal q_u is a singleton terminal class. The vertices of B are nonterminals. An $\mathcal{S}$ -path has endpoints in distinct terminal classes and no terminal in its interior. We seek exactly $|A|$ pairwise vertex-disjoint $\mathcal{S}$ -paths. Each such path uses two distinct terminals, and there are exactly $2|A|$ terminals; therefore a feasible solution uses every terminal. Feasibility is witnessed by the $|A|$ disjoint spokes $\{p_u, q_u\}$. If $A = \emptyset$, the unique required family is empty and the oracle is skipped; hence assume $|A| > 0$ when invoking the prescribed-positive-cardinality theorem.

Fix $0 \leq \rho \leq 2$; eventually $\rho = 16/13$. For notational economy, write u for the formal label underlying either p_u or a vertex of B . Give a physical edge uv unshifted weight

$$4d'(u, v) - \frac{\rho}{2}|\{u, v\} \cap B| - 2|\{u, v\} \cap A|. \quad (15)$$

Spokes have unshifted weight zero. Define the shifted weight of an auxiliary edge by adding two for each endpoint that is a terminal. The resulting physical-edge weights are

$$A-A : 4d', \quad A-B : 4d' - \rho/2, \quad B-B : 4d' - \rho.$$

All are nonnegative. Indeed, Lemma 2.9 says that a zero-distance pair can only be active-active; every active-outside or distinct outside-outside pair has $d' \geq 1$, and $0 \leq \rho \leq 2$. A spoke has shifted weight four, while a shadow edge has shifted weight at least three. Thus the exact nonnegative-length, prescribed-cardinality $\mathcal{S}$ -path theorem [19, Theorem 1 and Corollary 2] applies. Here and below polynomial time means polynomial bit complexity for rational input. Yamaguchi's reduction is over the rational field, and the rational-field implementation of Iwata-Kobayashi [9, Theorem 11.3] is polynomial in the encoding length of its matrix representation. The auxiliary weights here are rationals of polynomial encoding length (and $\rho = 16/13$ in the final algorithm), so the polynomial number of weight arithmetic operations also has polynomial bit complexity. We do not infer a strongly polynomial bit bound from an arithmetic-operation bound over an abstract field.

Lemma 3.4 (Exact objective). *The construction computes in polynomial time the maximum score*

$$W = a_5 + 2b_5 - \sum_{k=6}^{12} w_k c_k - 4E + \rho z, \quad (16)$$

where the chosen physical paths join distinct bags, have disjoint sets of explicit internal vertices in B , z is the number of those used outside vertices, and $E = \sum_P (\text{length}(P) - 1)$.

Proof. Every chosen path involving a shadow is a single edge: all shadow neighbors are terminals, and a terminal cannot be internal to an $\mathcal{S}$ -path. Hence every such path is either a spoke or a single shadow edge. Every remaining path has physical endpoints in two different bags and all of its explicit internal auxiliary vertices lie in B . Metric edges themselves remain atomic under the convention above. If a physical terminal is not an endpoint of such an interbag path, it must use its spoke. Consequently the shadows not used by spokes—precisely those corresponding to exposed ports—are paired by shadow edges within each bag. In particular, every bag has an even number of ports.

On any four vertices of a pentagon, the induced cycle edges form a four-vertex path and contain a perfect matching of two adjacent pairs. Thus four exposed ports have minimum shadow cost -2 , while two exposed ports have cost -1 exactly when they form a good pair. For $k \geq 6$, every even set of at least four shadows has a perfect matching that avoids all potentially bad edges. To prove this, observe that the potentially bad edges form a matching when k is even and a single k -cycle when k is odd; in either case their graph has maximum degree at most two and no triangle. Suppose a perfect matching of the exposed set contains a bad pair ab , and let cd be any other matched pair. At least one of $\{ac, bd\}$ and $\{ad, bc\}$ consists entirely of nonbad edges. Otherwise, choosing a bad edge from each alternative, together with ab , creates either a triangle or three bad edges incident with a or b . Replacing ab, cd by the all-nonbad alternative strictly decreases the number of bad matching edges. Iteration proves the assertion. Hence a positive minimum shadow penalty occurs only for a bad two-port state, and then it is w_k .

Let a physical $\mathcal{S}$ -path P have z_P internal vertices in B . Each of its two physical terminals contributes one -2 term in (15), and every internal B -vertex contributes two $-\rho/2$ terms. Therefore its unshifted weight is

$$4 \text{length}(P) - 4 - \rho z_P = 4(\text{length}(P) - 1) - \rho z_P.$$

The total unshifted shadow-edge weight is

$$-a_5 - 2b_5 + \sum_{k=6}^{12} w_k c_k,$$

and spokes contribute zero. Thus the total unshifted objective equals $-W$. Finally, every $\mathcal{S}$ -path has exactly two terminal endpoints, so shifting adds four per path, hence the same constant $4|A|$ to every feasible solution. Minimizing shifted weight is therefore exactly equivalent to maximizing (16). $\square$

3.5 Completing the outside vertices

An outside nontrivial even component with f_C edges and v_C vertices is *hit* if a chosen physical path contains at least one of its vertices as an explicit internal auxiliary vertex. Let $z_C > 0$ be the number of such vertices; they are distinct because the physical $\mathcal{S}$ -paths are vertex-disjoint. For a hit component, add the cheaper of the following two even subgraphs:

1. the entire component, of cost f_C ; or
2. two copies of a spanning forest having exactly one hit vertex in each tree, of cost $2(v_C - z_C)$.

For the second option, contract all z_C roots to one vertex, take a spanning tree of the connected quotient (discarding contraction loops), and then undo the contraction. The resulting graph has $v_C - z_C$ edges, is acyclic, and no component can contain two roots, since a path between two roots would contract to a cycle in the quotient tree. Roots unused by the tree simply become isolated one-root components. Doubling the forest makes all its degrees even, and every one of its components meets a chosen physical path. If the outside component is unhit, retain its original even edge set and reserve two units for one connector. An unused isolate is treated in the same way, with zero retained edges and a two-unit connector allowance.

Define

$$d_\rho(f_C, v_C) = \max \left\{ f_C + 2, \left(1 - \frac{\rho}{8}\right) f_C + \frac{\rho}{4} v_C \right\}. \quad (17)$$

Lemma 3.5 (Outside-component debit). *For $0 \leq \rho \leq 2$, an unhit component costs at most $d_\rho(f_C, v_C)$ including its connector allowance. For a hit component,*

$$\min\{f_C, 2(v_C - z_C)\} + \frac{\rho}{4} z_C \leq d_\rho(f_C, v_C). \quad (18)$$

A used or unused isolate has corresponding debit at most two.

Proof. The unhit cost is $f_C + 2$, the first branch of (17). For a hit component, consider

$$g(z) = \min\{f_C, 2(v_C - z)\} + \frac{\rho}{4} z \quad (0 \leq z \leq v_C).$$

If $f_C \leq 2v_C$, the two terms in the minimum cross at $z_0 = v_C - f_C/2$. The function is nondecreasing up to z_0 and nonincreasing after z_0 , because $\rho/4 \leq 1/2 < 2$. Hence

$$g(z) \leq g(z_0) = \left(1 - \frac{\rho}{8}\right) f_C + \frac{\rho}{4} v_C,$$

which proves (18). If $f_C > 2v_C$, the forest term is always smaller and

$$g(z) = 2v_C + \left(\frac{\rho}{4} - 2\right) z \leq 2v_C.$$

Moreover,

$$\left(1 - \frac{\rho}{8}\right) f_C + \frac{\rho}{4} v_C - 2v_C = \left(1 - \frac{\rho}{8}\right) (f_C - 2v_C) \geq 0,$$

so the same branch again bounds g . A used isolate contributes only its allocated reward term $\rho/4$, whereas an unused isolate needs a two-unit connector. Their maximum is $\max\{\rho/4, 2\} = 2$. $\square$

Proposition 3.6 (Deterministic paid bound). *The optimized paths and the switching algorithm produce a tour of cost*

$$P \leq \sum_{k=5}^{12} d_k r_k - \frac{W}{4} + \sum_C d_\rho(f_C, v_C) + 2i - 2. \quad (19)$$

Here the sum over C ranges over the unselected nontrivial components of the core.

Proof. Fix the physical paths selected by Lemma 3.4. Their total metric cost is

$$\sum_P \text{length}(P) = |\{P\}| + E.$$

The number of paths is half the total number of ports. Thus the j crossing units assigned to each $2j$ -port bag sum to $|\{P\}|$ exactly.

Start every multiport bag with its cheaper local cover and every two-port bag with its direct or rescued walk. Phase one changes no local cost. Equations (8)–(9) show that the incidence charges in (10)–(12) pay both the phase-two local-cost increase and two units for every final port-containing component. A zero-port bag contributes its cycle cost k and receives its own two-unit connector allowance. Therefore (13) and (14) give

$$\begin{aligned} & (\text{active local edges}) + (\text{physical paths}) + (\text{active connector allowances}) \\ & \leq \sum_{k=5}^{12} d_k r_k - \frac{1}{4}(a_5 + 2b_5) + \frac{1}{4} \sum_{k=6}^{12} w_k c_k + E \\ & = \sum_{k=5}^{12} d_k r_k - \frac{W}{4} + \frac{\rho}{4} z. \end{aligned} \tag{20}$$

The last equality is precisely the definition (16).

Partition the z used outside vertices among the hit nontrivial components and the used isolates. For every hit component, combine its share $\rho z_C/4$ from (20) with its cheaper completion; Lemma 3.5 bounds the result by $d_\rho(f_C, v_C)$. The same lemma covers an unhit component together with its two-unit connector allowance. Each isolate, used or unused, costs at most two after its share of $\rho z/4$ is included. This accounts for the sum over C and the term $2i$ in (19).

It remains only to justify the global subtraction of two. Before adding connectors, expand every metric edge into a shortest path in G and project all formal and private copies to their original vertices. By Lemma 2.9 and the rescue construction, the result is spanning and even, and projection can only merge components. Every component in the formal accounting has received a two-unit allowance: it is either a final port-containing component, a zero-port bag, an unhit outside component, or an unused isolate. Hit outside forests and used isolates already meet a physical-path component and create no additional component. If the number of accounted components is $m \geq 1$, the projected graph has at most m components. Lemma 2.2 therefore adds at most $2(m-1)$ original edges, whereas the accounting reserved $2m$. The resulting saving is exactly the final -2 in (19). The connected cyclic instances for which this construction is invoked are nonempty; the separately handled exact cases are recorded in Section 5.1. $\square$

4 Comparison with an optimum tour

Suppose initially there are at least two active bags. Let H' be the copied optimum metric Hamilton cycle of Lemma 2.9. Compress its segments between consecutive active vertices, retaining the original segment associated with each compressed edge. Segments with endpoints in the same bag are *internal*; the others are crossing. Here a segment consists of its formal Hamilton-cycle vertices and metric edges. Private internal occurrences that would arise when a metric edge is expanded in G are not segment vertices; their extra length is recorded by the metric excess defined below. At an active vertex, the internal degree is at most two. Moreover, the internal edges of one bag cannot contain a cycle: such a cycle would be the whole Hamilton cycle H' , contrary to the presence of a second active bag. Thus every bag's internal edges form a proper linear forest. Its nontrivial paths are *visits*, and its isolated vertices are *singleton visits*.

4.1 Main cycle and local transitions

Suppress the interior of each nontrivial visit with endpoints a, b and replace the resulting internal edge $\{p_a, p_b\}$ by the three-edge path p_a, q_a, q_b, p_b . Together with the compressed crossing edges, these replacement paths form one *main cycle*. A physical copy p_u of a singleton visit already lies on this cycle, but its shadow q_u does not. For every bag having at least one nontrivial visit, insert all its singleton shadows, in any order, into one or more of the shadow paths between visit endpoints; this does not change the endpoint pairing. If every visit of a bag is a singleton, use instead a separate shadow copy of its designated cycle. This separate cycle is disjoint from the main cycle.

During a modification in a bag with at least one nontrivial visit, delete its current shadow paths from the current main cycle. The remaining pieces pair the exposed shadow endpoints; call this perfect pairing R . A replacement shadow path cover is *legal* if the union of its endpoint pairing with R is one alternating cycle (a doubled edge when there is only one endpoint pair). This is equivalent to saying that reinserting the replacement paths produces one main cycle. We choose the replacement using this *current* R . Consequently the bags can be modified sequentially, and no simultaneous compatibility of independently chosen local transitions is assumed. Deleting local shadow paths here means temporarily removing their interior singleton-shadow vertices as well, while retaining their endpoints. Physical singleton vertices are never removed: they remain on the main cycle and may occur inside the outside trails defining R . Before a bag is first processed, its endpoint pairing is still the original visit pairing L , even if other bags have already been changed. Since the current main cycle is connected, $L \cup R$ is one alternating cycle. This is exactly the compatibility hypothesis used in the finite odd-cycle transition lemma.

Lemma 4.1 (Universal forbidden-edge bound). *Let there be $2s$ endpoints, paired by R , and m singleton vertices, where $s \geq 1$. If a forbidden graph on these vertices has maximum degree Δ , a legal shadow path cover has at most Δ forbidden edges. The exceptional domain $s = 1, m = 0$ needs at most one.*

Proof. Regard each R -pair as an orientable two-vertex block and each singleton as a one-vertex block. Order the $t = s + m$ blocks uniformly around a circle, orient every pair block independently, join consecutive entries, and then remove the R -edge inside every pair block. What remains is a shadow path cover, and restoring the removed R -edges recovers the single circular order; hence the cover is legal. For $t \geq 2$, let a, b, c count forbidden pairs joining two distinct pair blocks, a pair block to a singleton, and two singletons, respectively. Two prescribed blocks are adjacent with probability $2/(t - 1)$. Conditional on adjacency, a prescribed pair–pair edge is the connector with probability $1/4$, a prescribed pair–singleton edge with probability $1/2$, and a prescribed singleton–singleton edge with probability one. Therefore the expected number of forbidden cover edges is

$$\frac{a/2 + b + 2c}{t - 1}.$$

The same probabilities hold when $t = 2$, as is also seen directly from the two connector edges between the blocks. For $s \geq 2$, the degree inequalities $2a + b \leq 2s\Delta$ and $b + 2c \leq m\Delta$ imply

$$a/2 + b + 2c \leq \Delta(s/2 + m) - b/4 \leq \Delta(s + m - 1).$$

For $s = 1, m \geq 1$, $a = 0$ and the second degree inequality suffices. Thus some outcome has an integer number of forbidden edges no larger than the displayed expectation, and hence no larger than Δ . When $s = 1, m = 0$, the unique cover consists of one shadow edge, proving the stated exceptional bound. $\square$

Lemma 4.2 (Matching-forbidden transition). *If the forbidden pairs form a matching, a positive minimum forbidden count is possible only with one shadow path and at most four vertices: its endpoints form a forbidden pair, a singleton is forbidden to an endpoint, or its two singleton vertices form a forbidden pair.*

Proof. Take a legal cover of minimum forbidden-edge count and let uv be one of its forbidden edges. Suppose that a second cover edge ab is vertex-disjoint from uv . In the single cycle obtained by adjoining R , deleting uv and ab leaves two paths. Of the two crosswise reconnections

$$\{ua, vb\} \quad \text{and} \quad \{ub, va\},$$

exactly one rejoins those two paths into a single cycle. Use that reconnection and leave R fixed. The resulting cover is legal. Since the forbidden graph is a matching and uv is forbidden, all four cross-edges displayed above are good. The reconnection therefore strictly decreases the forbidden-edge count, a contradiction.

It follows that no cover edge can be vertex-disjoint from uv . A second shadow path would contain such an edge, so the cover has one path. A path on at least five vertices also has an edge disjoint from any specified edge. On a four-vertex path, only its middle edge has no disjoint path edge, and the middle edge joins its two internal vertices, which are the two singleton shadows. On a three-vertex path the only possible forbidden edge joins its internal singleton to an endpoint, and on a two-vertex path it joins the two endpoints. These are precisely the three alternatives in the statement. $\square$

4.2 Girth-sensitive transition bounds

For an internal compressed segment of a bag, call every vertex of B in the retained H' -segment a *return vertex* of that bag. The *internal metric excess* of the bag is the sum of $d'(h) - 1$ over the original H' -edges in its internal segments. This excess is a nonnegative integer: by Lemma 2.9, every zero-distance copied edge joins two different bags. Hence zero internal excess means that every original edge in an internal segment has unit length.

Lemma 4.3 (Even-cycle unit and one-return rescue). *For $k \in \{6, 8, 10, 12\}$, a bag with no positive original internal metric excess and at most one outside return vertex admits a legal shadow transition using no bad pair.*

Proof. For even k , the potentially bad pairs are the opposite pairs and hence form a matching. If every pair is nonbad there is nothing to prove, so apply Lemma 4.2. A positive minimum could occur only when there is one visit and the bag has, respectively, zero, one, or two singleton visits. We exclude its three exceptional patterns.

First assume that there is no return and that every internal unit edge is an edge of the designated cycle. A visit spanning the cycle is the cycle with one edge deleted, so its endpoints are adjacent, not opposite. If the visit misses one vertex x , it is the path obtained by deleting x , and its endpoints are the two neighbors of x ; neither forms an opposite pair with x . If it misses two vertices and the remaining vertices form one path, the two missing vertices must be adjacent. Thus none of the three exceptional forbidden pairs can occur.

Now suppose that there is exactly one return vertex z . Zero internal excess makes the corresponding part of the visit a two-edge path $u - z - v$, and all other original edges have unit length. We continue to assume temporarily that those other edges are cycle edges.

If there is no singleton, the visit itself is a spanning walk with k edges: it has $k - 2$ cycle edges and the two return edges. Hence its endpoint pair is nonbad because $k \leq h_k$.

Suppose next that x is the unique singleton. The cycle with x deleted is a path with $k - 2$ edges. The visit uses $k - 3$ of these cycle edges, so exactly one edge ab of that path is omitted. Those cycle edges form two (possibly one-vertex) paths, and the return $u - z - v$ joins one endpoint of each. It cannot join a to b , since ab together with $u - z - v$ would be a triangle. It cannot join the two cycle neighbors of x , since the two edges through x together with $u - z - v$ would be a four-cycle. It must therefore make a mixed join: one endpoint of the resulting visit is adjacent to x . If the other endpoint and x form the exceptional opposite pair, append the edge from the first endpoint to x . This gives a spanning walk between that pair of length $(k - 1) + 1 = k \leq h_k$. Thus the pair is nonbad. This argument includes the cases in which one of the two cycle pieces is a single vertex.

Finally, suppose that the two singleton vertices x, y are opposite; otherwise their exceptional pair is not forbidden. They are nonadjacent, and deleting them leaves two cycle paths with a total of $k - 4$ edges. The visit uses exactly these $k - 4$ cycle edges and the return. The return joins one endpoint of each path. It cannot join the two neighbors of x , or the two neighbors of y , because either choice creates a four-cycle through that singleton. The two free visit endpoints are therefore one neighbor of x and one neighbor of y . Append the two incident cycle edges to obtain a spanning x - y walk of length $(k - 2) + 2 = k \leq h_k$. Again the allegedly bad pair is nonbad.

It remains to justify the temporary cycle-edge assumption. Girth at least five forces any unit chord to have cyclic distance at least four. No such chord exists in C_6 . For $k = 8, 10, 12$, Lemma 4.4(b) gives, in the designated cycle together with any one such chord, a spanning walk of length at most h_k between every opposite pair. Consequently the presence of a unit chord makes every potentially bad pair nonbad, regardless of the visit decomposition. The preceding cases therefore exhaust all possibilities. $\square$

Lemma 4.4 (Bounded local verification). *The following finite assertions hold.*

- (a) *For $k = 7, 9, 11$, every proper nonempty cycle-edge subset, viewed as an internal visit forest, and every compatible outside pairing, admit a legal shadow path cover with at most one farthest-distance edge.*
- (b) *For $k = 8, 9, 10, 11, 12$, the cycle plus any unit chord of cyclic distance $4, \dots, \lfloor k/2 \rfloor$ has a spanning walk between every farthest pair of length at most h_k .*

Proof. Appendix A.1 proves part (a), and Appendix A.2 proves part (b). Rotation and reflection cover every chord placement; additional input edges can only shorten the certified walks. $\square$

Lemma 4.5 (Pentagon transition). *If a pentagon has zero internal metric excess and no return vertex, its shadow support contains at least one good edge. Here a good edge joins adjacent vertices of the designated pentagon.*

Proof. Girth five excludes every chord of the pentagon, so all internal unit edges are designated cycle edges. If the internal forest is nonempty, its component-size partition is one of

$$P_5; \quad P_4 + P_1; \quad P_3 + 2P_1; \quad P_2 + 3P_1; \quad P_3 + P_2; \quad 2P_2 + P_1.$$

This list is exhaustive because a proper linear forest on five cyclically ordered vertices has path components of sizes summing to five. Retain the original visit pairing, which is legal because its union with the current outside pairing R is one alternating cycle.

For P_5 , the two visit endpoints are adjacent. For $P_4 + P_1$, the singleton is adjacent to both visit endpoints and can be inserted between them. For $P_3 + 2P_1$, the two singleton vertices are adjacent; insert them consecutively in the shadow path. For $P_2 + 3P_1$, the three singletons lie

consecutively on the complementary four-edge arc between the visit endpoints, so insert them in that cyclic order. For $P_3 + P_2$, retain the shadow edge corresponding to the P_2 visit. For $2P_2 + P_1$, retain one P_2 shadow edge and insert the singleton into the other shadow path. In every case the indicated retained or inserted edge is good. If the internal forest is empty, all visits are singletons and the separate shadow pentagon consists of five good edges. $\square$

In the comparison below, a bag is called *contaminated* if it has positive internal metric excess or at least one return vertex. For a contaminated pentagon we conservatively claim no reward.

4.3 The perfect-matching point and all odd cuts

Let $X \subseteq A$ be the bag vertices incident with at least one crossing edge of the compressed H' . These are precisely the endpoints of nontrivial visits and the singleton visits; strict interiors of visits belong to $A \setminus X$. This use of X must not be confused with the full set A of activated bag vertices. After the legal transitions, the support consists of the main cycle and one separate shadow cycle for each all-singleton bag. It contains precisely the physical and shadow copies of vertices in X . Give each support-cycle edge weight $2/5$, and add weight $1/5$ on every spoke $\{p_u, q_u\}$ with $u \in X$. Give the spoke $\{p_u, q_u\}$ weight one when $u \in A \setminus X$. Thus an endpoint spoke that is itself an edge of the main cycle has total weight $3/5$, whereas a singleton spoke not on a support cycle has weight $1/5$. Every p_u and q_u with $u \in X$ is incident with two support-cycle edges and with its $1/5$ spoke layer; every strict visit interior has only its weight-one spoke. Hence every vertex has weighted degree one.

Lemma 4.6 (Fractional perfect-matching point). *This point belongs to the ordinary perfect-matching polytope.*

Proof. Nonnegativity and the degree equations were verified above. It remains to prove $x(\delta(S)) \geq 1$ for every odd vertex set S .

If S separates the ends of a spoke $\{p_u, q_u\}$ with $u \in A \setminus X$, that weight-one spoke proves the inequality. Otherwise, each such pair lies wholly inside or wholly outside S . Delete all these pairs from S and from its complement. This operation preserves the cut weight and the parity of $|S|$, so it reduces the problem to the vertices $\{p_u, q_u : u \in X\}$.

On these remaining vertices, the $1/5$ layers of the spokes form a perfect matching M_0 . Since $|S|$ is odd, $|M_0 \cap \delta(S)| \equiv |S| \pmod{2}$; in particular at least one baseline spoke crosses and contributes $1/5$. If S splits any support cycle, an even positive number of edges of that cycle crosses the cut, so the $2/5$ cycle layer contributes at least $4/5$. The required unit lower bound follows.

It remains to consider a shore S that is a union of whole support cycles. Contract every support cycle. Both ends of every baseline spoke lie on the main cycle, except in an all-singleton bag: there its physical ends lie on the main cycle and its shadow ends lie on that bag's separate shadow cycle. Thus the nonloop edges of M_0 form a star multigraph with the main cycle as center. The leaf corresponding to an all-singleton C_k has exactly $k \geq 5$ parallel spokes. If S separates any leaf from the center, those spokes contribute at least $5(1/5) = 1$. If no leaf is separated, S is either empty or the entire remaining vertex set, both of even cardinality; hence this last case cannot define an odd shore. All odd-cut inequalities hold. Edmonds's perfect-matching-polytope theorem [6] now supplies a convex decomposition into perfect matchings. $\square$

Every support edge joins distinct terminal classes: a physical support edge is a crossing edge between two bags, while a shadow edge or a spoke has at least one singleton shadow class as an endpoint. In each perfect matching, expand every selected physical crossing edge to its retained H' -segment. Distinct segments have distinct active endpoints and disjoint interiors in B , because

the compressed segments partition the Hamilton cycle between consecutive active vertices. The expanded edges are therefore pairwise vertex-disjoint physical $\mathcal{S}$ -paths. The selected shadow edges and spokes are single-edge $\mathcal{S}$ -paths covering all remaining terminals. Hence every perfect matching yields a feasible solution of the exact paid oracle.

For clarity, assign the following linear score to such a matching after its crossing edges have been expanded. A selected crossing segment P contributes

$$-4(\text{length}(P) - 1) + \rho z(P),$$

where $z(P)$ is its number of internal vertices in B . A selected adjacent pentagon shadow edge contributes $+1$; a selected bad C_k shadow edge contributes $-w_k$ for $k \geq 6$; all other shadow edges and all spokes contribute zero. This is an edge-linear quantity, so its expectation is determined by the fractional point.

There is a small but important distinction between this raw score and the canonical count expression in (16). An arbitrary matching in the convex decomposition need not minimize the shadow cost for its fixed exposed shadows. In every bag, replace its selected shadow matching by a minimum-cost perfect matching on the same exposed set. This changes no physical path, spoke, terminal, or port set and preserves feasibility. Lemma 3.4 shows that the resulting shadow score is exactly $a_5 + 2b_5 - \sum_{k=6}^{12} w_k c_k$ and is no smaller than the raw score. Thus every decomposed matching can be canonicalized without loss, and the oracle optimum W is at least the expected raw linear score of the comparison distribution.

4.4 Excess accounting and the complete comparison

For every compressed segment P , let

$$e(P) = \sum_{h \in E(P)} (d'(h) - 1), \quad z(P) = |V(P) \cap B|,$$

where the sum is over the original metric edges of H' retained inside P and the endpoints of P lie in A . Let e_X, e_I be the sums of $e(P)$ over crossing and internal segments, respectively, and let Z_X, Z_R be the corresponding sums of $z(P)$. The segments partition the edges of H' and their internal B -vertices partition B . Since H' has n' edges and length OPT , we have

$$e_X + e_I = e' = \text{OPT} - n', \quad Z_X + Z_R = |B|. \quad (21)$$

By Lemma 2.9, a zero-distance copied edge has its endpoints in different bags. Thus every internal original edge has length at least one, so $e_I \geq 0$ and each bag's internal excess is a nonnegative integer. In contrast, e_X may be negative. Define

$$\beta_\rho = \frac{8}{5} - \frac{2\rho}{5}. \quad (22)$$

Every compressed crossing edge is a main-cycle edge of fractional weight $2/5$, so its retained segment is selected with probability $2/5$. A segment P with $z(P) = z$ has $z + 1$ original edges and therefore

$$\text{length}(P) - 1 = e(P) + z.$$

Its expected contribution to the physical part of the raw score is

$$\frac{2}{5} \{-4(e(P) + z) + \rho z\} = -\frac{8}{5}e(P) - \beta_\rho z.$$

Summing over all crossing segments gives

$$-\frac{8}{5}e_X - \beta_\rho Z_X. \quad (23)$$

Every inserted zero edge has $e(P) = -1$ and $z(P) = 0$, and hence contributes $+8/5$; no special correction is being suppressed in (23).

We next bound the shadow part of the raw score. All-singleton bags need no transition: their separate shadow cycle uses only adjacent, hence nonbad, pairs. Consider a bag with a nontrivial visit.

For even k , the potentially bad pairs form a matching. By Lemma 4.1, the selected shadow support contains at most one bad edge. Its marginal is $2/5$ and $w_k = 4$, so the expected loss is at most $8/5$. Lemma 4.3 says more: if the bag's internal excess e_i is zero and its return count z_i is at most one, there is a transition containing no bad edge. Consequently an even-bag loss can occur only if $e_i \geq 1$ or $z_i \geq 2$.

For $k = 7, 9, 11$, the farthest-pair graph has maximum degree two. Lemma 4.1 therefore gives at most two potentially bad support edges, while Lemma 4.4(a) gives at most one when all internal unit edges are designated cycle edges and there is no return. If instead a zero-excess, zero-return bag contains a unit chord, no such chord is possible for $k = 7$, and Lemma 4.4(b) makes every farthest pair nonbad for $k = 9, 11$. Thus one bad edge is a universal baseline, and a second can occur only if $e_i \geq 1$ or $z_i \geq 1$. Since $w_k = 2$, both the baseline loss and the possible extra loss are $2(2/5) = 4/5$ in expectation.

Finally, Lemma 4.5 gives at least one good support edge in every pentagon with $e_i = z_i = 0$. Its marginal contribution is $2/5$. For a contaminated pentagon we discard this reward, losing at most $2/5$ relative to the claimed baseline.

For $\rho \leq 2$, $\beta_\rho \geq 4/5$. These observations give an injective bag-by-bag charge. If $e_i \geq 1$, any additional debit just listed is at most $8/5 \leq (8/5)e_i$. If $e_i = 0$, an even debit is at most $8/5$ and has $z_i \geq 2$, an odd debit is at most $4/5$ and has $z_i \geq 1$, and a pentagon debit is at most $2/5$ and has $z_i \geq 1$. In every case the debit is at most $\beta_\rho z_i$. Original internal segments belong to unique bags, so $\sum_i e_i = e_I$ and $\sum_i z_i = Z_R$; no unit of excess or return is charged twice. Hence the total additional internal debit is at most

$$\frac{8}{5}e_I + \beta_\rho Z_R. \quad (24)$$

This proves the following comparison.

Proposition 4.7 (Main score comparison). *If there are at least two active bags,*

$$W \geq \frac{2}{5}r_5 - \frac{4}{5}(r_7 + r_9 + r_{11}) - \frac{8}{5}(\text{OPT} - n') - \beta_\rho |B|. \quad (25)$$

Proof. The pentagon rewards and odd-cycle baseline penalties contribute at least

$$\frac{2}{5}r_5 - \frac{4}{5}(r_7 + r_9 + r_{11}).$$

Subtracting (24) and adding the crossing-segment ledger (23), the expected raw score is at least

$$\begin{aligned} & \frac{2}{5}r_5 - \frac{4}{5}(r_7 + r_9 + r_{11}) - \frac{8}{5}e_I - \beta_\rho Z_R - \frac{8}{5}e_X - \beta_\rho Z_X \\ &= \frac{2}{5}r_5 - \frac{4}{5}(r_7 + r_9 + r_{11}) - \frac{8}{5}(\text{OPT} - n') - \beta_\rho |B|, \end{aligned}$$

where the equality is (21). Canonicalizing every matching can only increase its score, and the exact oracle returns the best feasible score, so the same lower bound holds for W . Because no step assumed $e_X \geq 0$, the calculation includes the negative crossing excess created by whole-component zero copies. $\square$

5 Approximation guarantee and algorithm

5.1 Coefficient blend and main proof

Set $\rho = 16/13$, so $\beta_\rho = 72/65$. Combining (19) and (25) first gives

$$P \leq \frac{2}{5} \text{OPT} + \sum_{k=5}^{12} \left(d_k + \frac{\lambda_k}{4} \right) r_k - \frac{2}{5} n' + \frac{\beta_\rho}{4} |B| + \sum_C d_\rho(f_C, v_C) + 2i - 2.$$

Here $\lambda_5 = -2/5$, $\lambda_k = 4/5$ for $k \in \{7, 9, 11\}$, and $\lambda_k = 0$ for even k , exactly as in Table 1. Now use

$$n' = \sum_{k=5}^{12} N_k + v + i, \quad |B| = v + i, \quad v = \sum_C v_C, \quad r_k = \frac{N_k}{k}.$$

Since

$$\frac{\beta_\rho}{4} - \frac{2}{5} = -\frac{\rho}{10}, \quad p_k = \frac{d_k + \lambda_k/4}{k} - \frac{2}{5},$$

collecting terms yields

$$P \leq \frac{2}{5} \text{OPT} + \sum_{k=5}^{12} p_k N_k + \sum_C \left(d_\rho(f_C, v_C) - \frac{\rho}{10} v_C \right) + \left(2 - \frac{\rho}{10} \right) i - 2. \quad (26)$$

Lemma 5.1 (Outside debit). *Every remaining nontrivial component satisfies*

$$d_{16/13}(f_C, v_C) - \frac{8}{65} v_C \leq \frac{67}{65} f_C.$$

Proof. The left side is

$$\max \left\{ f_C + 2 - \frac{8}{65} v_C, \frac{11}{13} f_C + \frac{12}{65} v_C \right\}.$$

For the second branch, $v_C \leq f_C$ gives

$$\frac{11}{13} f_C + \frac{12}{65} v_C \leq \frac{55 + 12}{65} f_C = \frac{67}{65} f_C.$$

For the first branch, multiplication by 65 shows that the desired inequality is equivalent to

$$f_C + 4v_C \geq 65.$$

A remaining cycle has $f_C = v_C \geq 13$, and hence equality or better. A remaining noncycle has $f_C \geq 18$ and $f_C \leq 2v_C - 6$, so

$$f_C + 4v_C \geq 3f_C + 12 \geq 66.$$

Both branches are therefore bounded as claimed. $\square$

Summing Lemma 5.1 over the remaining nontrivial components, and substituting the exact p_k values from Table 1, turns (26) into the first bound below. Equation (4) is the second:

$$P \leq \frac{2}{5} \text{OPT} + \frac{49}{50} N_5 + \frac{14}{15} N_6 + \frac{69}{70} N_7 + \frac{39}{40} N_8 + \frac{91}{90} N_9 \\ + N_{10} + \frac{113}{110} N_{11} + \frac{61}{60} N_{12} + \frac{67}{65} f + \frac{122}{65} i - 2, \quad (27)$$

$$D \leq \frac{7}{5} N_5 + \frac{4}{3} N_6 + \frac{9}{7} N_7 + \frac{5}{4} N_8 + \frac{11}{9} N_9 + \frac{6}{5} N_{10} + \frac{13}{11} N_{11} + \frac{7}{6} N_{12} + \frac{15}{13} f + 2i - 2. \quad (28)$$

The optimum is not needed to construct either tour; it occurs only in the analysis of the first line.

Proposition 5.2 (Exact blend). *Assume that G contains a cycle and activation produces at least two bags. Then*

$$\begin{aligned} \min\{P, D\} \leq & \frac{101}{73} \text{OPT} - 2 - \frac{11}{219} N_6 - \frac{8}{511} N_7 - \frac{9}{292} N_8 - \frac{4}{657} N_9 \\ & - \frac{7}{365} N_{10} - \frac{5}{438} N_{12} - \frac{2}{949} f - \frac{200}{949} i. \end{aligned} \quad (29)$$

Proof. The weights $60/73$ and $13/73$ are nonnegative and sum to one, so

$$\min\{P, D\} \leq \frac{60P + 13D}{73}.$$

In this convex combination, the explicit coefficient of OPT is $(60/73)(2/5) = 24/73$. The coefficients of N_5 and N_{11} both equal $77/73$. Subtracting every other N_k - and f -coefficient from $77/73$, and the isolate coefficient from $2(77/73) = 154/73$, gives

variable	N_6	N_7	N_8	N_9	N_{10}	N_{12}	f	i
deficit	$11/219$	$8/511$	$9/292$	$4/657$	$7/365$	$5/438$	$2/949$	$200/949$

For example,

$$\frac{60}{73} \frac{49}{50} + \frac{13}{73} \frac{7}{5} = \frac{77}{73}, \quad \frac{60}{73} \frac{113}{110} + \frac{13}{73} \frac{13}{11} = \frac{77}{73}.$$

Consequently, the weighted average is at most

$$\begin{aligned} & \frac{24}{73} \text{OPT} + \frac{77}{73} \left(\sum_{k=5}^{12} N_k + f + 2i \right) - 2 \\ & - \frac{11}{219} N_6 - \frac{8}{511} N_7 - \frac{9}{292} N_8 - \frac{4}{657} N_9 - \frac{7}{365} N_{10} - \frac{5}{438} N_{12} - \frac{2}{949} f - \frac{200}{949} i. \end{aligned}$$

By (3), the parenthesized quantity is $\Phi \leq \text{OPT}$. Replacing it by OPT gives $(24 + 77) \text{OPT} / 73 = 101 \text{OPT} / 73$ and proves (29) with exactly the displayed deficits and the common subtraction of two. $\square$

Small cases and the uniform factor. A tree is solved exactly by doubling all its edges: every edge is a bridge and must be traversed at least twice. This includes the one-vertex tree, for which both the doubled edge set and the optimum have length zero.

Lemma 5.3 (At most one active bag). *Suppose that G contains a cycle. If activation produces at most one bag, the direct tour satisfies*

$$D \leq \frac{15}{13} \Phi \leq \frac{15}{13} \text{OPT}.$$

If there is no bag, the stronger bound $D \leq (15/13)\Phi - 2$ holds.

Proof. If there is no bag, then (4) and $\Phi = f + 2i$ give

$$D \leq \frac{15}{13} f + 2i - 2 = \Phi + \frac{2}{13} f - 2 \leq \frac{15}{13} \Phi - 2.$$

If there is exactly one bag, of length k , then $N_k = k$ and all other N_j vanish. Again by (4),

$$\begin{aligned} D & \leq (k + 2) + \frac{15}{13} f + 2i - 2 \\ & = \Phi + \frac{2}{13} f \leq \frac{15}{13} \Phi, \end{aligned}$$

because now $\Phi = k + f + 2i$. The inequality $\Phi \leq \text{OPT}$ is Lemma 2.5; it applies because G is connected and cyclic. Both displayed final inequalities use only the immediate bound $f \leq \Phi$. The variable i was retained throughout, so outside isolates are already included in both cases. $\square$

For additional perspective, if G itself is one cycle, returning that cycle is exact: its length is $|V|$, while every Hamilton cycle in its shortest-path metric has $|V|$ positive-length edges.

The additive refinement (29) is not claimed for every exact small case; for example, a single pentagon has optimum five and does not satisfy the refined right-hand side $(101/73) \cdot 5 - 2 = 359/73 < 5$.

Proof of Theorem 1.1. If G is a tree, the doubled-tree branch is optimal. Otherwise G contains a cycle and Lemma 2.5 applies. With at most one active bag, Lemma 5.3 proves the claimed ratio because

$$\frac{15}{13} < \frac{101}{73}.$$

With at least two active bags, both tours P, D are constructed and Proposition 5.2 gives $\min\{P, D\} \leq 101 \text{OPT} / 73$ (indeed, with the stated additive slack). These cases exhaust every connected input of girth at least five. The polynomial runtime is verified next. $\square$

5.2 Explicit algorithm and runtime

1. Verify that the input is a finite connected simple graph and contains no triangle or four-cycle. If a test fails, report that the theorem is not applicable. If the input is a tree, return two copies of every edge.
2. Compute the lexicographically minimum binary even core and construct its doubled-quotient tour D .
3. Activate whole components as in Section 2.3; decompose them into cycles, create the formal copies, and compute the formal shortest-path pseudometric. If at most one bag is created, return D , as justified by Lemma 5.3.
4. For every potentially bad farthest pair in every bag, run the full-input product-graph search of Section 3 and store a shortest rescue of length at most h_k when one exists. Construct the physical/shadow $\mathcal{S}$ -path instance with $\rho = 16/13$, request exactly $|A|$ vertex-disjoint $\mathcal{S}$ -paths, and solve the resulting minimum-length prescribed-cardinality problem.
5. Decode each bag using its cheaper local cover, or its precomputed two-port rescue walk. Perform the free phase and then the repeated free/paid switching phase of Section 3.
6. Complete each outside component using the cheaper of its whole even subgraph and a doubled rooted forest. Keep unhit components and unused isolates for quotient completion.
7. Expand every atomic metric edge into an original shortest path, using private internal occurrences as specified in Section 2.3; then identify all formal and private copies with their original vertices, erase the resulting zero loops, and add a doubled spanning tree of the component quotient. This gives the tour P . Return the cheaper of P and D ; compute an Euler tour of the returned multigraph, and shortcut it only if a metric Hamilton cycle rather than a graphic closed walk is desired.

All steps are polynomial, including in bit complexity. The core-matching auxiliary graph is polynomial, $|A| \leq |F| \leq |E|$, and the complete physical layer is at most quadratic. There are $O(|A|)$ rescue tests, each with at most $2^{12}|V|$ states and $O(2^{12}|E|)$ transitions. At $\rho = 16/13$, multiplying the nonnegative shifted weights by thirteen gives integers of polynomial encoding length, so Yamaguchi's reduction and deterministic weighted linear matroid parity solve Step 4 in polynomial time [19, 9]. The local rules have constant size, and every accepted switch lowers the component count, so only polynomially many are accepted. All remaining operations are standard graph algorithms. The optimum tour is used only in the analysis, never as algorithmic input.

A Human-readable audit of the finite local claims

This appendix proves the two finite assertions in Lemma 4.4 without relying on executable search. The displayed witness tables are certificates whose validity can be checked directly.

A.1 Odd-cycle unit transitions

Let $k \in \{7, 9, 11\}$ and put $q = (k - 1)/2$. Number the cycle vertices modulo k . Two vertices are at farthest cyclic distance precisely when their difference is q or $q + 1 = -q \pmod{k}$. Hence the farthest-pair graph

$$H_k = \{\{x, x + q\} : x \in \mathbb{Z}_k\}$$

is itself one chordless k -cycle. Indeed, every vertex has the two neighbors $x \pm q$; moreover, $\gcd(k, q) = \gcd(k, (k - 1)/2) = 1$, so this 2-regular graph is connected.

Let the internal visit forest have $s \geq 1$ nontrivial paths, and let U be its set of u singleton vertices. Its $2s$ endpoints are paired by the internal visits. A compatible outside pairing R is a perfect pairing of these endpoints whose union with the visit pairing is one alternating cycle; in particular, R has s pairs. Regard every pair of R as an orientable two-vertex block and every member of U as a singleton block. Put the blocks in a cyclic order, join consecutive entries, and remove the edge inside every R -block. The remaining edges form a spanning shadow path cover with the prescribed endpoints. Restoring the removed R -edges recovers the single cyclic order, so the cover is legal. Conversely, in the cycle obtained by adjoining R to any legal cover, the endpoints of each R -edge are consecutive; regarding those edges as blocks recovers an order of the stated kind.

Suppose first that $s \geq 2$, and put $t = s + u$. Let α, β, γ be the numbers of edges of H_k joining, respectively, two distinct pair blocks, a pair block to a singleton, and two singletons. Order the t blocks uniformly on a circle and orient pair blocks independently. A specified pair–pair, pair–singleton, or singleton–singleton edge becomes a connector with probability, respectively, $1/(2(t - 1))$, $1/(t - 1)$, or $2/(t - 1)$. These formulas also give the correct expected multiplicity when $t = 2$. By linearity of expectation, the expected number X of farthest shadow edges is therefore

$$\mathbb{E}X = \frac{\alpha/2 + \beta + 2\gamma}{s + u - 1}. \quad (30)$$

The $2s$ vertices in pair blocks have total H_k -degree at most $4s$, and the u singleton vertices have total H_k -degree at most $2u$. Discarding edges internal to an R -block can only weaken the first count, so

$$2\alpha + \beta \leq 4s, \quad \beta + 2\gamma \leq 2u.$$

The first inequality gives $\alpha/2 \leq s - \beta/4$ and the second gives $2\gamma \leq 2u - \beta$. Adding these bounds proves

$$\alpha/2 + \beta + 2\gamma \leq s + 2u - \beta/4. \quad (31)$$

If $s \geq 3$, the right side is strictly smaller than $2(s + u - 1)$, so $\mathbb{E}X < 2$.

If $s = 2$ and $\beta > 0$, the last inequality is again strict relative to $2(s + u - 1) = 2(u + 1)$. If $\beta = 0$, the four pair-block vertices form a proper vertex subset of the chordless cycle H_k and hence induce a forest with at most three edges. Thus $\alpha \leq 3$, and

$$\alpha/2 + \beta + 2\gamma \leq 3/2 + 2u < 2(u + 1).$$

In every case $\mathbb{E}X < 2$. Since X is a nonnegative integer, some block order has $X \leq 1$.

It remains to treat $s = 1$. By rotation and reversal, the unique internal visit is the path $0, 1, \dots, r$, where $1 \leq r \leq k - 1$; its singleton set is $\{r + 1, \dots, k - 1\}$. Table 2 gives a shadow path from 0 to r through every singleton. Each row is a permutation of $\{0, r, r + 1, \dots, k - 1\}$, with 0 first and r last. In every row, consecutive labels differ modulo k by neither q nor $q + 1$. Thus none of its edges belongs to H_k , which is stronger than required.

Table 2: One-visit shadow paths. In each block the order starts at 0, visits every singleton exactly once, and ends at r .

$k = 7$		$k = 9$		$k = 11$	
r	order	r	order	r	order
1	0, 2, 3, 4, 5, 6, 1	1	0, 2, 3, 4, 5, 6, 7, 8, 1	1	0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 1
2	0, 5, 6, 4, 3, 2	2	0, 3, 4, 5, 6, 7, 8, 2	2	0, 3, 4, 5, 6, 7, 8, 9, 10, 2
3	0, 5, 6, 4, 3	3	0, 6, 4, 7, 8, 5, 3	3	0, 4, 5, 6, 7, 8, 9, 10, 3
4	0, 5, 6, 4	4	0, 6, 5, 8, 7, 4	4	0, 7, 5, 6, 9, 10, 8, 4
5	0, 6, 5	5	0, 6, 7, 8, 5	5	0, 7, 6, 8, 10, 9, 5
6	0, 6	6	0, 7, 8, 6	6	0, 7, 8, 9, 10, 6
		7	0, 8, 7	7	0, 8, 9, 10, 7
		8	0, 8	8	0, 9, 10, 8
				9	0, 10, 9
				10	0, 10

A nonempty proper edge subset of a cycle is a linear forest with at least one nontrivial path, so $s \geq 1$. The cases $s \geq 2$ and $s = 1$ therefore exhaust every state in Lemma 4.4(a). This proves that part of the lemma without profile enumeration.

A.2 Chord-rescue walks

Normalize the cycle to $0, 1, \dots, k - 1, 0$ and the chord to $0d$, where $4 \leq d \leq \lfloor k/2 \rfloor$. (The lower bound on d is exactly what girth at least five requires after replacing d by $k - d$ if necessary.) For a fixed start a , consider the finite state graph

$$\mathcal{Q}_{k,d} = \{(v, S) : v \in S \subseteq \mathbb{Z}_k\}.$$

It has an arc of length one

$$(v, S) \longrightarrow (w, S \cup \{w\}) \quad \text{whenever} \quad vw \in E(C_k) \cup \{0d\}. \quad (32)$$

Every a - b walk $v_0, \dots, v_\ell$ in the chorded cycle yields a directed state-graph walk by taking $S_i = \{v_0, \dots, v_i\}$; it ends at $(b, \mathbb{Z}_k)$ exactly when the original walk is spanning. Conversely, projecting any directed state-graph walk onto its first coordinate gives a walk in the chorded cycle, while the update rule in (32) says that its second coordinate is precisely the set of vertices visited so far. The distance from $(a, \{a\})$ to $(b, \mathbb{Z}_k)$ is therefore exactly the minimum length of a spanning a - b walk.

Let $\mathcal{F}_k$ denote the unordered farthest pairs: for even k they are the $k/2$ opposite pairs, and for odd k they are the k pairs $\{x, x + (k - 1)/2\}$. The reflection $\sigma(x) = d - x \pmod{k}$ preserves all

cycle edges and interchanges the ends of the chord $0d$. It therefore acts as an involution on $\mathcal{F}_k$. Table 3 displays proposed spanning walks for orbit representatives; the audit immediately after the table proves that there is exactly one row for each orbit. A witness for a reflected pair is obtained by applying σ and, if necessary, reversing the displayed walk.

Table 3: Chord-rescue witnesses. Each comma-separated sequence is a spanning walk; repeated labels are intentional.

k	d	pair	walk
8	4	0, 4	0, 1, 2, 3, 4, 0, 7, 6, 5, 4
8	4	1, 5	1, 2, 3, 4, 0, 7, 6, 5
8	4	2, 6	2, 1, 0, 4, 3, 4, 5, 6, 7, 6
9	4	0, 4	0, 1, 2, 3, 4, 0, 8, 7, 6, 5, 4
9	4	0, 5	0, 1, 2, 3, 4, 0, 8, 7, 6, 5
9	4	1, 5	1, 2, 3, 4, 0, 8, 7, 6, 5
9	4	1, 6	1, 2, 3, 4, 0, 8, 7, 6, 5, 6
9	4	2, 6	2, 1, 2, 3, 4, 0, 8, 7, 6, 5, 6
10	4	0, 5	0, 1, 2, 3, 4, 0, 9, 8, 7, 6, 5
10	4	1, 6	1, 2, 3, 4, 0, 9, 8, 7, 6, 5, 6
10	4	2, 7	2, 1, 0, 4, 3, 4, 5, 6, 7, 8, 9, 8, 7
10	5	0, 5	0, 1, 2, 3, 4, 5, 0, 9, 8, 7, 6, 5
10	5	1, 6	1, 2, 3, 4, 5, 0, 9, 8, 7, 6
10	5	2, 7	2, 1, 2, 3, 4, 5, 0, 9, 8, 7, 6, 7
k	d	pair	walk
11	4	0, 5	0, 1, 2, 3, 4, 0, 10, 9, 8, 7, 6, 5
11	4	0, 6	0, 1, 2, 3, 4, 0, 10, 9, 8, 7, 6, 5, 6
11	4	1, 6	1, 2, 3, 4, 0, 10, 9, 8, 7, 6, 5, 6
11	4	1, 7	1, 2, 3, 4, 0, 10, 9, 8, 7, 6, 5, 6, 7
11	4	2, 7	2, 1, 2, 3, 4, 0, 10, 9, 8, 7, 6, 5, 6, 7
11	4	5, 10	5, 4, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
11	5	0, 5	0, 1, 2, 3, 4, 5, 0, 10, 9, 8, 7, 6, 5
11	5	0, 6	0, 1, 2, 3, 4, 5, 0, 10, 9, 8, 7, 6
11	5	1, 6	1, 2, 3, 4, 5, 0, 10, 9, 8, 7, 6
11	5	1, 7	1, 2, 3, 4, 5, 0, 10, 9, 8, 7, 6, 7
11	5	2, 7	2, 1, 2, 3, 4, 5, 0, 10, 9, 8, 7, 6, 7
11	5	2, 8	2, 1, 2, 3, 4, 5, 0, 10, 9, 8, 7, 6, 7, 8
k	d	pair	walk
12	4	0, 6	0, 1, 2, 3, 4, 0, 11, 10, 9, 8, 7, 6, 5, 6
12	4	1, 7	1, 2, 3, 4, 0, 11, 10, 9, 8, 7, 6, 5, 6, 7
12	4	2, 8	2, 1, 0, 4, 3, 4, 5, 6, 7, 8, 9, 10, 11, 10, 9, 8
12	4	5, 11	5, 4, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11
12	5	0, 6	0, 1, 2, 3, 4, 5, 0, 11, 10, 9, 8, 7, 6
12	5	1, 7	1, 2, 3, 4, 5, 0, 11, 10, 9, 8, 7, 6, 7
12	5	2, 8	2, 1, 2, 3, 4, 5, 0, 11, 10, 9, 8, 7, 6, 7, 8
12	6	0, 6	0, 1, 2, 3, 4, 5, 6, 0, 11, 10, 9, 8, 7, 6
12	6	1, 7	1, 2, 3, 4, 5, 6, 0, 11, 10, 9, 8, 7
12	6	2, 8	2, 1, 2, 3, 4, 5, 6, 0, 11, 10, 9, 8, 7, 8
12	6	3, 9	3, 2, 1, 0, 6, 5, 4, 5, 6, 7, 8, 9, 10, 11, 10, 9

Here is a separate audit of both orbit coverage and length. Because σ is an involution, Burnside's lemma gives

$$\#(\mathcal{F}_k / \langle \sigma \rangle) = \frac{|\mathcal{F}_k| + |\text{Fix}_{\mathcal{F}_k}(\sigma)|}{2}.$$

Solving $\sigma(\{a, b\}) = \{a, b\}$ gives the fixed pairs in the third column below. The resulting orbit count equals the number of displayed rows in every (k, d) group; direct application of σ to the listed representatives also shows that those rows belong to distinct orbits.

k	d	$ \mathcal{F}_k $	fixed pairs	orbits	rows	maximum length	h_k
8	4	4	$\{0, 4\}, \{2, 6\}$	3	3	9	9
9	4	9	$\{0, 4\}$	5	5	10	10
10	4	5	$\{2, 7\}$	3	3	12	12
10	5	5	$\{0, 5\}$	3	3	11	12
11	4	11	$\{5, 10\}$	6	6	13	13
11	5	11	$\{0, 5\}$	6	6	13	13
12	4	6	$\{2, 8\}, \{5, 11\}$	4	4	15	15
12	5	6	none	3	3	14	15
12	6	6	$\{0, 6\}, \{3, 9\}$	4	4	15	15

Finally, each displayed sequence starts and ends at its stated pair, contains every label in $\mathbb{Z}_k$, and has every consecutive pair either differing by one modulo k or equal to the chord pair $\{0, d\}$. The maximum-length column counts the number of separators in the longest sequence of each group. Thus every listed object is a spanning walk of at most h_k edges, and the orbit calculation covers every farthest pair. This proves Lemma 4.4(b).

Use of AI: We prepared a list of proof strategies and techniques related to approximating TSP. We fed this to multiple AI tools (Claude, ChatGPT and two open-weight models with open-source harnesses) and prompted these AI tools to improve the existing approximation bounds. When an AI tool was hallucinating, stuck, or going in the wrong direction, we stopped that AI tool, corrected the intermediate .md files, added more specific new ideas and switched to a different AI tool with more detailed fresh prompts. Equations written by one AI tool were critiqued, verified and expanded by a completely different AI tool in an iterative manner, till we were satisfied with their correctness and details. The author is solely responsible for the correctness of the proofs.

References

- [1] S. Boyd, R. Sitters, S. van der Ster, and L. Stougie. The traveling salesman problem on cubic and subcubic graphs. *Mathematical Programming* 144 (2014), 227–245. [arXiv:1107.1052](#).
- [2] N. Christofides. Worst-case analysis of a new heuristic for the travelling salesman problem. Technical Report 388, Graduate School of Industrial Administration, Carnegie Mellon University, 1976.
- [3] A. Çivril. 4/3-Approximation of Graphic TSP. Withdrawn manuscript, arXiv:2305.05411v9 (2026). [arXiv:2305.05411](#).
- [4] J. Correa, O. Larré, and J. A. Soto. TSP tours in cubic graphs: beyond 4/3. *SIAM Journal on Discrete Mathematics* 29(2) (2015), 915–939. [doi:10.1137/140972925](#).
- [5] Z. Dvořák, D. Král, and B. Mohar. Graphic TSP in cubic graphs. *STACS 2017*, LIPIcs 66, 27:1–27:13. [doi:10.4230/LIPIcs.STACS.2017.27](#).
- [6] J. Edmonds. Maximum matching and a polyhedron with 0,1-vertices. *Journal of Research of the National Bureau of Standards, Section B* 69B (1965), 125–130. [doi:10.6028/jres.069B.013](#).
- [7] F. Frei, A. Ghazy, T. A. Hartmann, F. Hörsch, and D. Marx. From Chinese postman to salesman and beyond II: inapproximability and parameterized complexity. *SIAM Journal on Discrete Mathematics* 40(3) (2026), 1066–1106. [doi:10.1137/25M1736402](#); [arXiv:2502.18541](#).
- [8] H. N. Gabow. Data structures for weighted matching and extensions to b -matching and f -factors. *ACM Transactions on Algorithms* 14(3) (2018), Article 39. [doi:10.1145/3183369](#); [arXiv:1611.07541](#).
- [9] S. Iwata and Y. Kobayashi. A weighted linear matroid parity algorithm. *SIAM Journal on Computing* 51(2) (2022), STOC17-238–STOC17-280. [doi:10.1137/17M1141709](#). [arXiv:1905.13371](#).
- [10] A. R. Karlin, N. Klein, and S. Oveis Gharan. A (slightly) improved approximation algorithm for metric TSP. *Operations Research* 72(6) (2024), 2543–2594. [doi:10.1287/opre.2022.2338](#).

- [11] T. Mömke and O. Svensson. Approximating graphic TSP by matchings. *FOCS* (2011), 560–569. [arXiv:1104.3090](#). Journal development: Removing and adding edges for the traveling salesman problem, *Journal of the ACM* 63 (2016), Article 2.
- [12] M. Mucha. 13/9-approximation for Graphic TSP. *Theory of Computing Systems* 55(4) (2014), 640–657. [doi:10.1007/s00224-012-9439-7](#); [arXiv:1108.1130](#).
- [13] A. Newman. An improved analysis of the Mömke–Svensson algorithm for graph-TSP on subquartic graphs. *SIAM Journal on Discrete Mathematics* 34 (2020), 865–884. [arXiv:1407.2524](#), journal version v4.
- [14] S. Oveis Gharan, A. Saberi, and M. Singh. A randomized rounding approach to the traveling salesman problem. *FOCS 2011*, 550–559. [doi:10.1109/FOCS.2011.80](#).
- [15] A. Sebő and J. Vygen. Shorter tours by nicer ears: 7/5-approximation for the graph-TSP, 3/2 for the path version, and 4/3 for two-edge-connected subgraphs. *Combinatorica* 34 (2014), 597–629. [arXiv:1201.1870](#); [doi:10.1007/s00493-011-2960-3](#).
- [16] A. I. Serdyukov. Some extremal bypasses in graphs. *Upravlyaemye Sistemy* 17 (1978), 76–79. [In Russian.]
- [17] V. Traub and J. Vygen. *Approximation Algorithms for Traveling Salesman Problems*. Cambridge University Press, 2024. [doi:10.1017/9781009445436](#). Author manuscript: <https://www.or.uni-bonn.de/tspbook/book.pdf>.
- [18] M. C. Wigal, Y. Yoo, and X. Yu. Approximating TSP walks in subcubic graphs. *Journal of Combinatorial Theory, Series B* 158 (2023), 70–104. [doi:10.1016/j.jctb.2022.09.002](#); [arXiv:2112.06278](#).
- [19] Y. Yamaguchi. Shortest disjoint $\mathcal{S}$ -paths via weighted linear matroid parity. *ISAAC 2016*, LIPIcs 64, 63:1–63:13. [doi:10.4230/LIPIcs.ISAAC.2016.63](#).